SNAPSMACK OPERABILITY AUDIT — DING DONG BELL

OPAUDIT 015 — One photograph, three places, three different comment counts

Audit ID	2026-09-14-015
Report date	2026-09-14
First observed	2026-09-14, the day after OPAUDIT 013 was called watched. Sean opened the same photograph (“Got my eye on you”, allinthewrist.photoblogs.fyi) in three places and screenshotted all three.
Theme	“Comments federate” had been checked in one direction, one hop, one pair of servers. A thread is not one hop. Every place that shows the photograph must show the same conversation, or the feature is a rumour.
Surfaces	The public comment thread on the blog; the same post on Pixelfed; the same post inside another SnapSmack blog’s FEDIVERSE reader; the ActivityPub Note the blog publishes.
State	FIXED — CONFIRMING Three fixes built and deployed from 0.7.712D onward; the three-place live comparison, especially Pixelfed accepting a forwarded reply, is still not recorded as watched.
Watched live?	**No.** The defect was watched (three screenshots, three counts). The fix has not been.
Reporter	Sean (looked at the photograph as three different readers would) + Claude (traced the three causes, wrote the three fixes).
Related	OPAUDIT 013 (replies stored, never shown — this is its sequel: shown, but not the same set everywhere); OPAUDIT 014 (cron — ruled out here: none of the three gaps was a delivery problem).
Publication	Public. No exploitable detail.

1. SUMMARY

The same photograph had four comments on the blog that owns it, two on Pixelfed, and none in another SnapSmack blog’s reader. Three separate causes, none of them the delivery queue: the reader only knew how to ask a Pixelfed-style API for a thread, so a SnapSmack post always answered with nothing; a comment typed on the blog before the outbound fix of 708D had simply never been sent; and a reply that came from a third server was kept by the blog but never passed on to the blog’s followers, so Pixelfed had no way to know it existed.

2. WHAT A HUMAN SAW

Where	Comments shown
allinthewrist.photoblogs.fyi (the source)	4 — zuckit@unzucked.ca, mccormickphoto@pixelfed.social, The Dude, Cthulhu

pixelfed.social	2 — mccormickphoto (its own), Cthulhu (the 708D mirror, watched the day before)
unzucked.ca FEDIVERSE reader	0 — not even zuckit's own reply, sent from that very reader

3. ROOT CAUSE — THREE OF THEM

The reader (0 comments). The reader's thread loader took the post's address, pulled the last number out of it, and asked `/api/v1/statuses/N/context` on that host — the Mastodon/Pixelfed API. A SnapSmack blog has no such API. The request 404'd and the thread rendered empty, for every SnapSmack post, since the reader was written.

The Dude (missing on Pixelfed). A local comment, older than Cthulhu's. 708D mirrors each *new* community comment into the classic table and federates it; everything typed before 708D existed sat in the community table only. Cthulhu was typed after the deploy and went out; The Dude was typed before and never would.

zuckit's reply (missing on Pixelfed). unzucked.ca sent its reply to allinthewrist, which stored and displayed it. Pixelfed follows allinthewrist, not unzucked. ActivityPub's answer to this is inbox forwarding (spec §7.1.2): the server that owns the post forwards replies it receives to its own followers. Mastodon and Pixelfed do this. SnapSmack did not. On top of that, a SnapSmack blog receiving a forwarded activity rejected it outright, because the activity's actor did not match the server that signed the delivery — correct for a forgery, wrong for a forward.

4. EVIDENCE AT DIAGNOSIS

Three screenshots of the same photograph, counts as above. allinthewrist's delivery log: queue empty, worker firing every ten minutes (the cron fix of OPAUDIT 014 was already in), so delivery was ruled out before code was read. The reader's thread function (`sv_status_replies`) contained one code path, the Mastodon API; the inbox handler's first check was actor-equals-signer-or-401; the 708D mirror lived only in the new-comment handler.

5. WHAT WE DID (0.7.712D)

A standard `replies` collection on every post. Each content Note now carries `replies` pointing at `/ap/note/{i|p|l}/N/replies` : an ordered collection of the approved thread — remote replies listed by their own addresses (the fetcher dereferences them from where they were written; the blog never vouches for someone else's words), local comments embedded in full. Mastodon reads this collection when a thread is opened. The reader now falls back to it when the origin has no Mastodon API, following `first` when the collection is paged and dereferencing bare ids. Same output shape, so the reader's page code did not change.

Inbox forwarding. A public reply that reaches one of the blog's own posts is forwarded, as received, to every follower inbox except the replier's host — and only when the blog kept it (a reply held for moderation is not amplified). On the receiving side a Create whose actor is not the signer is accepted only after the

object is fetched from its origin and is attributed to the claimed actor on that actor's own host; anything else is the same 401 as before. The receipt line says "forwarded by X — object verified from its origin".

One-time backfill. The delivery cron, once per version, mirrors every visible community comment that has no local mirror yet and federates it with its original date; a second pass finds nothing to do. Push & Tools gets a SEND UNSENT BLOG COMMENTS button that runs the same routine now.

6. VERIFIED — AND WHAT REMAINS

Regression `tests/fediverse-thread-whole-everywhere-regression.php` (19 checks) and the existing fediverse suites pass. Nothing has been watched live. To close this: deploy 712D to allinthewrist and unzucked; open the photograph in unzucked's reader and count four; wait one delivery tick and count The Dude on pixelfed.social; have zuckit reply again and see whether pixelfed.social shows it. The last of those is the one that can still fail for reasons outside our code: a receiving server may drop a forwarded activity it cannot verify its own way. If Pixelfed does, the honest state of forwarding becomes *contained* — it works between SnapSmack blogs and Mastodon, and Pixelfed needs a different route — and this entry says so.

7. EYEBALL NEXT

The three counts, after deploy. Then: hiding or deleting a comment does not yet retract it from the fediverse (708D's known gap, still open). And the same three-places check for likes, which OPAUDIT 009 left undiagnosed.

8. THE DOCTRINE THIS INCIDENT CREATED

> *A federated feature is watched when the same thing is seen from every place it can be looked at — not when one hop between one pair of servers succeeds once. Count it in three places before calling it working.*